Chapter 2: Working with Images

Comprehensive Theory, Practical Procedures, Tools & Solved Board Questions (Session 2026-2027)

1. Introduction to Graphics in LibreOffice Writer

Visual elements significantly improve the clarity, readability, and overall impact of a document. In LibreOffice Writer, graphics fall under two primary categories:

- **Pictures / Image Files:** Bitmap or vector graphics created outside Writer using external tools (e.g., scanners, cameras, graphic design software) or stored on disk in formats such as *JPEG, PNG, GIF, BMP*.
- **Drawing Objects:** Ready-made vector shapes created directly within LibreOffice Writer using the *Drawing Toolbar*, including *Lines, Curves, Basic AutoShapes, Graphic TextArt (Fontwork), and Text Boxes*.

> **File Formats Note [CBSE Board PYQ]: GIF** (Graphics Interchange Format), **JPEG / JPG** (Joint Photographic Experts Group), and **PNG** (Portable Network Graphics) are examples of standard *Graphic / Image file formats*.

2. Methods of Inserting Images into a Document

LibreOffice Writer provides four primary methods to insert an image into a document:

Method A: Using the Insert Image Dialog

1. Click in the document at the position where the image should appear.
2. Select **Insert > Image...** from the Menu bar (or click the *Insert Image* icon on the Standard toolbar).
3. In the *Insert Image* dialog box, browse your computer and select the desired file.
4. Click the **Open** button. By default, the image is embedded and placed centered above the clicked paragraph.

Method B: Inserting Images by Linking (Embedding vs Linking)

When you insert an image normally, it is **embedded** (stored completely inside the document file). If a document contains many high-resolution photos, the document file size increases substantially. To keep the file size small, the image should be **linked** instead.

1. Select **Insert > Image...** from the Menu bar.
2. In the *Insert Image* dialog, locate and select the image file.
3. Check the **Link** checkbox at the bottom of the dialog box before clicking **Open**.
4. Click **Open**. Writer saves only the reference/file path of the image. The image is retrieved dynamically from disk each time the document is opened.

> **Linking via Drag & Drop [CBSE Board 2024]:** When dragging an image from File Explorer into Writer, holding down `Ctrl + Shift` while dropping creates a **link** instead of embedding it.

Method C: Using Drag and Drop

1. Open LibreOffice Writer alongside your operating system's File Explorer.
2. Select the image file in the folder window.
3. Drag the file directly onto the Writer document canvas and release the mouse button at the target position.

Method D: Using Copy and Paste / From Clipboard

1. In File Explorer or an image viewer, right-click the image and select **Copy** (or press `Ctrl + C`).
2. Click in the Writer document at the desired location, right-click and choose **Paste** (or press `Ctrl + V`).

Method E: Inserting Images from the Built-in Gallery

1. Open the Gallery by selecting **Tools > Gallery** (or clicking the *Gallery* icon on the Standard toolbar / Sidebar).
2. Browse through the available themes (e.g., Arrows, Diagrams, Icons, Shapes).
3. Select the desired graphic and drag-and-drop it into the document, or right-click the graphic and select **Insert**.

3. Modifying Images: Properties, Resizing & Cropping

When an image is selected in Writer, green sizing handles appear around its border, and the **Image Toolbar** and **Properties Deck** become active (accessible via `Ctrl + F5` or **Format > Image > Properties**).

Key Image Visual Properties

- **Brightness:** Controls the overall lightness or darkness of the image.
- **Contrast:** Adjusts the difference in luminance and tone between light and dark regions.
- **Color Mode:** Specifies how color channels are rendered:
  - *Default:* Unchanged original color spectrum.
  - *Grayscale:* Converts all colors into shades of gray (256 tones).
  - *Black/White:* Converts the image strictly into high-contrast monochrome pixels.
  - *Watermark:* Significantly reduces contrast and increases brightness so the image can sit behind readable text.
- **Transparency:** Varies image opacity from **0%** (completely opaque) to **100%** (completely invisible). Useful for custom watermark effects.

Resizing an Image (Scaling)

1. Click the image to display its eight sizing handles.
2. To resize interactively: Click and drag any corner sizing handle. To preserve proportions (prevent distortion), hold down the `Shift` key while dragging.
3. To resize precisely: Open the *Properties* pane or right-click > **Properties...** > **Type** tab. Specify exact Width and Height, and check **Keep ratio**.

> **Raster vs Vector Resizing [CBSE Board 2024]:** Resizing a **Raster image** (bitmap formats like JPEG/PNG) degrades visual quality and resolution (pixelation), whereas a **Vector image** scales mathematically without loss of sharpness.

Cropping an Image

Cropping refers to removing unwanted peripheral sections from an image starting from its outer boundaries. It does not scale or shrink the contents; it trims away the outside edges.

1. Click the image to select it.
2. Right-click on the image and choose **Crop** from the context menu (or click the *Crop* icon on the Image toolbar).
3. Crop handles (black corner and side markers) appear around the image boundary.
4. Position the cursor over a crop handle until it turns into a cropping tool icon, then drag inward to remove unwanted areas.

Keep Scale (Cropping Option) [CBSE 2023]	Keep Image Size (Cropping Option) [CBSE 2023]
Cropping cuts off the unwanted border while the remaining part of the image **maintains its original magnification and scale**.	Cropping removes the unwanted border, but the remaining visible portion is **automatically enlarged or reduced to fit the original image frame size**.
The overall physical dimensions of the image on the page decrease.	The overall physical boundary dimensions of the image frame remain unchanged on the page.

4. Graphic Image Filters (Image Toolbar)

The **Filter** icon on the Image toolbar (or menu path **Format > Image > Filter**) provides 11 built-in graphical effects:

Filter Name	Function / Effect on Image	Board Exam Relevance
Invert	Reverses the color values of a colored image or brightness values of a grayscale image (photographic negative effect).	CBSE Set-4 PYQ
Sharpen	Increases contrast between adjacent pixels, making edges and details crisper.	CBSE SQP & Board 2023
Smooth	Softens contrast and blends adjacent pixels together to reduce high-frequency noise.	CBSE Objective Option
Remove Noise	Eliminates isolated speckles (stray single pixels) across the image surface.	Image Cleaning Tool
Solarisation	Mimics the effect of exposing photographic film to light during development.	Lighting Filter
Aging	Simulates the appearance of an old photograph by converting colors and applying sepia/brown tones.	Stylistic Effect
Posterize	Reduces the total number of colors to create a flat, painting-like look.	Artistic Reduction
Pop Art	Converts the image into a high-saturation, stylized pop-art graphic.	Modern Art Effect
Charcoal Sketch	Transforms the photograph into a hand-drawn black-and-white charcoal illustration.	Drawing Effect
Relief	Calculates light source shadows to create an embossed, 3D sculpted appearance.	Emboss Effect
Mosaic	Groups neighboring clusters of pixels together into solid rectangular tiles.	Pixelation Effect

5. Working with Drawing Objects & Grouping

Activating the Drawing Toolbar

If the Drawing toolbar is not visible at the bottom of the Writer window, activate it using either:

- Select **View > Toolbars > Drawing** from the Menu bar.
- Click the **Show Draw Functions** icon on the Standard toolbar.

Drawing Shapes and Setting Properties

- **Drawing Shapes:** Click the desired shape icon (e.g., *Rectangle, Ellipse, Line, Basic Shapes*) and click-and-drag in the document. Hold down `Shift` while dragging to create perfect squares or circles.
- **Formatting Lines & Arrows:** Select the shape > choose **Format > Text Box and Shape > Line...** (or right-click > **Line...**). Customize line style, line color, width (thickness), and arrow styles for either end.
- **Formatting Area Fill:** Select the shape > choose **Format > Text Box and Shape > Area...** (or right-click > **Area...**). Select fill type: *None, Color, Gradient, Hatching, Bitmap, or Pattern*.
- **Rotating Shapes:** Select the drawing shape, right-click and choose **Rotate**. The corner square handles transform into red circular rotation handles. Drag any corner handle to rotate clockwise or counter-clockwise.

Grouping Drawing Objects [CBSE Board 2024]

Grouping combines multiple independent drawing shapes so that they behave as a single entity. Once grouped, moving, scaling, deleting, or applying transformations modifies all objects together.
Step-by-Step Procedure to Group Objects:

1. Click the first drawing object to select it.
2. Hold down the `Shift` key on the keyboard and click each subsequent drawing object you wish to include in the group.
3. With all target objects selected, choose **Format > Group > Group** from the Menu bar (or right-click the selection and choose **Group > Group**).
4. To reverse the action later, select the group and choose **Format > Group > Ungroup**.

6. Positioning Graphics in Text: The Big Four Settings

Precise placement of an image relative to text and other elements is governed by four distinct controls:

A. Text Wrapping [CBSE Board 2023, 2024 Case Study]

Text wrapping dictates how surrounding document prose flows around, across, or behind a graphic.

- **Wrap Off (None):** Text stops above the top edge of the graphic and resumes below the bottom edge. No text appears alongside either side.
- **Page Wrap (Parallel):** Text flows smoothly around all four sides of the graphic frame.
- **Optimal Page Wrap:** The default intelligent wrapping mode. Text flows around the graphic; however, if the space between the image and margin is less than **2 cm**, Writer leaves that side blank to avoid fragmented single-word lines.
- **Wrap Left (Before):** Text flows exclusively on the left side of the graphic.
- **Wrap Right (After):** Text flows exclusively on the right side of the graphic.
- **Wrap Through:** Superimposes the image on top of the text. Text flows behind the graphic, requiring image transparency adjustment to remain readable.
- **In Background:** Pushes the graphic completely behind the text layer, acting as a background illustration.

B. Anchoring [CBSE Board 2023, SQP 2024]

Anchoring establishes the anchor point or reference datum for the graphic. As text is edited or added, the graphic maintains its relationship to this anchor point.

- **To Page:** The graphic remains fixed at an exact physical coordinate on a specific page, irrespective of text flow or added paragraphs.
- **To Paragraph (Default):** The graphic is tied to a specific paragraph. If the paragraph moves to another page or changes position, the graphic moves with it.
- **To Character:** The graphic is tied to a specific character. It moves with that character but remains positioned in its own separate frame.
- **As Character:** The graphic is treated as an inline text glyph. Its height alters the line height of the sentence, and it flows identically to typed words. Ideal for small math symbols or inline icons.

C. Alignment

Determines the horizontal and vertical positioning of the image relative to its anchor point:

- **Horizontal Alignment:** Left, Centered, Right.
- **Vertical Alignment:** Top, Center, Bottom.

D. Arrangement (Stacking Order) [CBSE Board 2023, SQP Case Study]

Controls the z-order (stacking sequence) when multiple overlapping images or drawing shapes share the same space:

- **Bring to Front:** Moves the selected object to the absolute top of the stack, in front of all overlapping elements.
- **Forward One:** Moves the selected object one layer up in the stacking hierarchy.
- **Back One:** Moves the selected object one layer down in the stacking hierarchy.
- **Send to Back:** Moves the selected object to the absolute bottom of the stack, behind all overlapping graphics.

7. Solved Board Examination & Case Study Bank

Q1. Differentiate between "Keep scale" and "Keep image size" options while cropping an image. [2 Marks - CBSE Board 2023]

Ans: In **Keep scale**, cropping trims away the outer boundary of the image while the scale and magnification of the remaining image stay constant, resulting in a smaller overall image on the page. In **Keep image size**, cropping removes the outer borders, but the remaining image is scaled/enlarged to fit the original image frame size.

Q2. What do you understand by Anchoring in Word Processor? [2 Marks - CBSE SQP]

Ans: Anchoring acts as the reference point for the horizontal and vertical placement of an image or drawing object. It determines whether the graphic stays fixed to a specific page coordinate (*To Page*), moves alongside a paragraph (*To Paragraph*), or behaves as an inline text element (*As Character*).

Q3. Write the main steps to group drawing objects in LibreOffice Writer. [2 Marks - CBSE Board 2024]

Ans:

1. Click on the first drawing object to select it.
2. Hold down the `Shift` key and click each additional drawing object to be grouped.
3. Choose **Format > Group > Group** from the Menu bar (or right-click and select *Group > Group*).

Q4. How do the "Wrap Off" and "Wrap Through" text wrapping options differ when placing images alongside text in Writer? [2 Marks - Board Paper Set-4]

Ans: In **Wrap Off**, text stops above the graphic and resumes below it, leaving both left and right sides completely blank. In **Wrap Through**, the graphic is superimposed directly over the text layer, allowing text to continue behind the image.

Q5. CASE STUDY: An editor designing an e-magazine faces four formatting challenges. Answer the following questions: [4 Marks - CBSE Sample Paper 2024-25 / 2025-26]

(a) Which feature should the team use to solve the problem of overlapping images?
Ans: **Arrangement** (or *Bring to Front / Send to Back / Forward One / Back One*).
(b) Which setting will help the editor keep the main image fixed in the center of the cover page no matter how much text is added or removed?
Ans: **Anchoring > To Page** (along with Centered Alignment).
(c) Which positioning option will make text flow smoothly around the images in the article?
Ans: **Text Wrapping** (e.g., *Page Wrap / Optimal Page Wrap*).
(d) Which arrangement tool should be used to ensure the advertisement image appears above all other elements?
Ans: **Bring to Front**.

Q6. Gaurav noticed that dragging and dropping an image increases the document file size significantly. Suggest a way to link the image instead. What is the key benefit of linking? [2 Marks - CBSE Board 2024]

Ans: While dragging and dropping, Gaurav should hold down the `Ctrl + Shift` keys simultaneously before releasing the mouse button, or select *Insert > Image* and check the **Link** box. The primary benefit is that it keeps the document file size small because the image data is stored externally on disk and loaded via its file path reference only when the document is opened.